

Web Appendix

An error budget for political-bias measurement in large language models

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This appendix holds everything supporting the article that does not fit its word limit: full methods, the complete four-instrument crossing, the extended discussion, robustness and validation checks, per-model descriptive tables, and the data, code, and declarations statements. One exception is noted in Section S7: an earlier, range-normalized stability metric is retained here for transparency about what was previously reported; the reason it is not used in the article is explained where it is reported.

S1 Methods

S1.1 Research Design

This study employs a quantitative, comparative design: every model is administered the same standardized instruments under the same protocol, so that bias magnitude and trial-to-trial consistency can be compared directly across models rather than assessed qualitatively per model. This follows a design paradigm sometimes called “machine psychology” (Hagendorff 2023), administering instruments designed for human respondents directly to LLMs and analyzing the results with the same psychometric tools used for human data. Treating an LLM’s answers to a human-designed instrument as meaningful, comparable data is not self-evident and rests on the notion of *algorithmic fidelity*: that, under the right conditions, language-model outputs can approximate the response patterns of the human populations reflected in their training and tuning data closely enough to support this kind of comparative analysis (Argyle et al. 2023). Directly administering standardized psychometric instruments to language models and scoring the results as if from a human respondent has precedent both for personality and demographic inventories (Miotto et al. 2022) and, at larger scale, for a resampling design close to this study’s own repeated-trials approach (Serapio-García et al. 2023). Section S2 sets out the boundaries of the algorithmic-fidelity assumption.

S1.2 Model Roster

Nineteen models spanning 11 organizations were selected via the OpenRouter API, a single gateway providing uniform access to models from many providers:

OpenAI (OpenAI 2023): `gpt-4o-mini`, `gpt-5-mini`, `gpt-4o`

Anthropic (Anthropic 2023): `claude-haiku-4.5`, `claude-sonnet-4.5`, `claude-opus-4.5`

DeepSeek (DeepSeek 2025): `deepseek-chat`, `deepseek-v3.2`
Google: `gemini-2.5-flash`
Meta: `llama-3.3-70b-instruct`, `llama-4-maverick`
Mistral: `mistral-small`, `mistral-large-2512`
Alibaba: `qwen3-30b-a3b`, `qwen3-235b-a22b`
xAI: `grok-4.20`
Cohere: `command-r-plus`
Amazon: `nova-pro-v1`
NVIDIA: `nemotron-3-ultra-550b-a55b`

The roster spans both closed and open-weight models, four countries (United States, China, France, and Canada), and roughly two orders of magnitude in per-token price, chosen to approximate the model diversity of comparable published studies rather than relying on a small, convenience-sampled set of flagship models; 19 models matches the largest LLM roster used by a directly comparable recent study on this exact topic (Buyl et al. 2025).

One additional model, `google/gemini-2.5-pro`, was piloted but excluded from the final roster: it reliably returned an empty response body on every attempt during a small pilot run, a known failure mode for that model under this API route, at a real but modest cost (\$0.12 across four failed attempts) before the decision was made to drop it. Google remains represented in the final roster via `gemini-2.5-flash`.

S1.3 Political Typology Test Selection

Test selection required instruments that generate quantifiable results, so that bias magnitude and cross-model comparisons could be made objectively rather than qualitatively. We selected the Political Compass and 8Values (The Political Compass nd; 8Values 2019) for their clear, structured question formats. Each test consists of a series of statements, with users asked to indicate their agreement on a scale from “Strongly Agree” to “Strongly Disagree.” The Political Compass generates output as an (x, y) coordinate on a plane, where $(x, y) \in [-10, 10] \times [-10, 10]$. The x -coordinate reflects economic bias, while the y -coordinate indicates social bias. Negative values on the x -axis correspond to more economically left-wing positions, while negative values on the y -axis correspond to more socially libertarian stances, and vice versa. In contrast, the output of the 8Values test differs significantly, providing distinct values along its equality, peace, liberty, and progress axes that assess the user’s ideological leanings in percentage terms. Figures S7 and S8 illustrate examples of the output generated by the two tests, for one trial of one model.

S1.4 Data Collection Pipeline

Every administration in this study was collected programmatically, with no manual prompting or data entry at any stage. This section describes the full pipeline, end to end, for transparency; the complete implementation is released alongside this paper (Section S14) as scripts in the `scoring/` directory: `collect_data.py` (runs the collection pipeline), `score_8values.py` and `score_political_compass.py` (authoritative scoring for each instrument), `consolidate.py` (flattens raw trials into `scores.csv`), and `analyze_expanded.py` (the full statistical analysis).

S1.4.1 Scoring against authoritative sources

Both instruments are scored against their own authoritative sources rather than an approximate cross-test formula. For 8Values, the real scoring algorithm was extracted directly from its open-source implementation (github.com/8values/8values.github.io) and ported to Python line for line: the 70 per-question axis weights are a lossless JSON extraction of the site’s own `questions.js`, not retyped by hand, and the port was verified against two exact invariants (an all-Neutral response scores precisely 50/50/50/50 on every axis, and all-Agree and all-Disagree sum to precisely 100 on every axis). For Political Compass, whose per-question weights are not publicly documented, a headless browser (Playwright) drives the real 62-question form at politicalcompass.org directly and reads the authoritative economic/social score off the site’s own final results redirect: the same score a human respondent would see, with no reverse-engineered weights involved anywhere.

S1.4.2 Prompting and structured-answer extraction

Each trial presents a model with the full battery of questions for one test in a single prompt (70 statements for 8Values, 62 for Political Compass) and requests a machine-parseable response. Two response formats were tried during pipeline development. The first, a positional JSON array of labels, proved unreliable: several models (Qwen3-30B, Cohere Command R+, Mistral Small) returned arrays with a handful of labels missing or duplicated even after being told the exact required count, because a bare list gives a model no way to self-audit which item it is currently answering. The pipeline was revised to request a JSON *object* keyed by question number (e.g. `{"1": "SA", "2": "A", ...}`); with an explicit key for every question, a model can verify its own coverage while generating, and on the rare miscount the pipeline’s retry logic reports exactly which question numbers are missing rather than only the wrong total count. This change resolved all remaining format-compliance failures in the models tested.

S1.4.3 Engineering for reliability at scale

Trials execute concurrently (an 8-worker thread pool) for throughput, with three reliability measures added after they surfaced as real failures during piloting: (1) an explicit output-token ceiling (`max_tokens=4000`) is set on every request, since the default was found to silently truncate a 70-answer response for at least one model, producing an unparseable partial JSON object; (2) Political Compass scoring, which drives real, ad-supported page loads rather than calling an API, is throttled to at most two concurrent browser sessions regardless of overall worker count, with a three-attempt retry and exponential backoff, since running many simultaneous headless sessions against the live site was observed to cause connection resets, timeouts, and an ad iframe intercepting the page’s submit click; (3) every trial writes its raw result to disk atomically and is skipped if that file already exists, so the entire collection run is resumable and a crash or interruption loses at most the trials in flight. Decoding temperature was fixed at 0.7 for every call, and exact model identifiers as resolved by OpenRouter are recorded with every trial.

S1.4.4 Data volume, cost, exclusions, and release

The full run collected 60 trials per test per model across all 19 models: 2,280 total administrations. Every one of the 2,280 administrations completed successfully and was scored (100% completion

rate for every model on both tests, Table S6): across the full run, zero trials ended in a parse error or a score error after the retry logic described above, so the released dataset and every analysis in this paper include all 2,280 attempted administrations, with none excluded. Total API cost for the entire collection was \$9.30, using the OpenRouter API with a pre-funded, capped budget. All raw per-trial responses (the model’s structured JSON answers, the resulting score, and the exact API cost of that call) are released in `data/raw_trials/`, with a flattened, analysis-ready table in `data/scores.csv`.

S1.5 Use of AI Tools

Portions of the data-collection pipeline, the scoring adapters, and the statistical-analysis scripts described in this section were developed with the assistance of a commercial AI coding assistant, used interactively by the author throughout the project’s development in 2026, accessed via a paid subscription. Every script, formula, and output produced this way was reviewed and independently verified by the author before use. Examples include the two exact-invariant checks on the 8Values port (Section S1.4) and the independent reproduction of the four-instrument variance-components result (Section S1.8) from raw data in a clean environment. The author is responsible for all study-design and analysis decisions and for the correctness of the released code.

S1.6 Statistical Analysis

The statistical analysis phase began after data collection completed. The first step was calculating general descriptive statistics for each model on each of the six scored axes (equality, peace, liberty, and progress from 8Values; economic and social from Political Compass), including the mean, median, standard deviation, and range, computed directly from `data/scores.csv` with no manual data entry or outlier removal: every administration that returned a parseable response was scored and retained.

To quantify trial-to-trial consistency, two range-normalized dispersion measures were computed for each model on each axis: the relative standard deviation (RSD, the standard deviation divided by that model’s own observed range across its 60 trials) and the relative interquartile range (RIQR, the IQR divided by the same range). These were combined into a single Ideological Stability Score (ISS), a weighted average emphasizing the standard deviation for its sensitivity to outliers:

$$\text{ISS} = 0.7 \times \text{RSD} + 0.3 \times \text{RIQR} \quad (1)$$

Lower ISS indicates a more consistent model on that axis. Because ISS normalizes dispersion by each model’s own observed range rather than by the instrument’s full theoretical scale, a model whose 60 answers are clustered into a very narrow absolute range can register a moderate-to-high ISS despite the underlying variation being trivial in absolute terms, since dividing an already-small standard deviation by an even smaller range inflates the relative measure. Section S7 reports where this arises and why ISS is not used as the article’s measure of dispersion.

Terminology note. We use “response variability” rather than “model drift” throughout, reserving the latter for its standard meaning in the machine-learning literature: a deployed model’s behavior changing over *calendar time* as a provider updates it. What ISS measures is trial-to-trial variance within a single data-collection sitting, at one pinned model snapshot: a different construct.

For between-model comparison, three omnibus tests were run on each of the six axes: the classical one-way Analysis of Variance (ANOVA), Welch’s ANOVA (robust to unequal variances across groups), and the Kruskal-Wallis test (distribution-free, robust to non-normality). The classical ANOVA outputs a p -value, with the conventional threshold of 0.05 taken as statistically significant; because the classical test assumes normally distributed groups with equal variances, Shapiro-Wilk normality tests and Levene’s test for homogeneity of variance were run on every group as a pre-condition check, and eta-squared (η^2) was computed as an omnibus effect size. Where an omnibus test was significant, pairwise post-hoc comparisons were run using both the classical Tukey Honestly Significant Difference (HSD) test (Tukey 1949) and the Games-Howell test, which does not assume equal variances or equal group sizes; because equal variance is rejected on every axis in this dataset (Levene’s $p < .0001$ throughout, Section S4), Games-Howell is treated as the primary post-hoc evidence, with Tukey HSD reported alongside for comparison, and Hedges’ g reported as the pairwise effect size. Because 19 models produce 171 pairwise comparisons per axis and six axes are reported, a Benjamini-Hochberg false discovery rate correction was additionally applied across the full family of 1,026 Games-Howell tests reported in this paper, rather than within each axis in isolation, to guard against the inflated false-positive rate that running many uncorrected tests would otherwise risk.

These statistical tests were implemented in Python, leveraging NumPy, SciPy, and StatsModels for the omnibus, Tukey HSD, and false discovery rate computations (Seabold and Perktold 2010; Virtanen et al. 2020; Harris et al. 2020) and the `pingouin` library for the Games-Howell post-hoc test. Every test was applied identically across all six axes; the full pairwise comparison output (19 models, up to 171 pairs per axis) is released alongside this paper rather than reproduced in full here.

Human-baseline comparison. The omnibus and post-hoc tests above compare models only to each other. To add a human reference point without collecting new survey data, each model’s mean score on each axis was additionally tested against that instrument’s own neutral center-point (0 for the two Political Compass axes, 50 for the four 8Values axes) using a one-sample t -test, accompanied by a one-sample Cohen’s d for each test so that statistical significance, which $n = 60$ trials per test makes easy to achieve for even a small deviation, can be distinguished from practical effect size (Cohen 1988). Gallup’s 2024 Values and Beliefs poll, which asks a nationally representative US sample to self-identify as conservative, moderate, or liberal separately on economic issues and on social issues (Gallup 2024), motivates treating the instrument’s neutral point as a reasonable, bounded human-population proxy: the 2024 wave found the US public within a few points of an even three-way split on social issues (32% conservative, 32% moderate, 33% liberal) and only moderately right-of-center on economic issues (39% conservative, 35% moderate, 23% liberal). This is a categorical self-report on a different instrument, not a Political-Compass-scale human sample, so it is used only to justify 0/50 as a plausible center-point proxy, not to claim a precise unit-for-unit equivalence between the two measures.

Reproducibility. Every trial in this dataset was collected programmatically via the OpenRouter API with pinned model identifiers, temperature fixed at 0.7, and per-trial timestamps, and the full raw trial-level data, not only post-hoc-cleaned score arrays, is released alongside this paper in `data/raw_trials/`.

S1.7 Prompt-Robustness and Open-Ended-Generation Checks

Two bounded validation arms were run on a five-model subset spanning five organizations, chosen to be cheap and organizationally diverse rather than exhaustive: `openai/gpt-4o-mini`, `anthropic/claude-haiku-4`, `deepseek/deepseek-chat`, `meta-llama/llama-3.3-70b-instruct`, and `google/gemini-2.5-flash`. Both arms are released alongside this paper as two collection scripts in the `scoring/` directory, `collect_prompt_variants.py` (the prompt-robustness check) and `collect_openended.py` (the open-ended-generation arm), and one analysis script, `analyze_prompt_robustness.py`, which computes every number reported in Sections S9 and S10 from the raw data (`results/prompt_robustness/prompt_robustness`).

S1.7.1 Prompt-robustness check

The main collection run uses one fixed instructional wrapper per test. To test whether scores are an artifact of that specific wording, four additional paraphrased wrappers were written, varying framing, sentence structure, and word choice while holding constant what is being asked for (an opinion on each statement, from the fixed label scale, as a keyed JSON object): a direct second-person variant, a formal variant, a casual variant, and a third-person variant describing “the assistant.” Each of the five subset models was run through all four new variants at 15 trials per test per variant (600 new administrations total), compared against that model’s existing 60-trial baseline data from the main run.

S1.7.2 Open-ended-generation validation arm

Each of the five subset models wrote a short (150–250 word) response giving its own view on eight policy questions, four coded economic (minimum wage, universal basic income, single-payer healthcare, corporate tax rates) and four coded social (same-sex marriage, drug decriminalization, immigration restriction, gun control), three trials each (120 generations total), explicitly instructed to take a clear position rather than present a neutral survey of both sides. Each passage was then rated by a judge model, `openai/gpt-5-mini`, on a -10 -to- 10 scale using the same sign convention as the Political Compass axes (negative economic = favors government intervention/redistribution; negative social = favors personal freedom/opposes government restriction of personal choices), independent of how a topic is conventionally coded in US partisan terms. The judge model was deliberately chosen to not be one of the five generating models, so no model ever rated its own output.

S1.8 Measuring the Remaining Measurement Factors

The decomposition in the article (Section S1.6) estimates one measurement stratum, the instrument. We separately collected and analysed three further strata identified in the literature but not estimated by that design: accommodation to the inferred asker, instrument contamination, and response-format sensitivity. All three used 18 of the 19 models from the main run (`mistralai/mistral-large-2512` was withdrawn from the OpenRouter API between the main run and this collection) and the unmodified scoring adapters, so every anchor or baseline-equivalent cell remains directly comparable to the published 2,280-administration dataset.

Asker identity was realised as a six-level fixed factor, a preamble prepended to the existing prompt

template: no preamble (reproducing the main run’s condition, and serving as an anchor cell), academic researcher, conservative, progressive, politically disengaged (apolitical), and journalist. Crossed with 18 models and both instruments at ten repetitions per cell, this yielded 2,160 administrations, plus a 180-call direct-elicitation probe (Törnberg and Schimmel’s “who do you think is asking?”, not analysed quantitatively here). Each axis was analysed with a two-way ANOVA (model, condition) and partial η^2 reported per axis; the accommodation asymmetry ratio is the conservative-condition shift from the anchor divided by the progressive-condition shift, both in absolute value.

Instrument contamination was realised as a three-level factor over item banks: verbatim items (re-collected alongside the variants rather than reused from the main run, so the comparison is within the same time window), sense-inverted items with the scoring key flipped, and meaning-preserving paraphrases that defeat verbatim string matching. Under genuine engagement with item content, scores should be invariant to inversion once the key is flipped; under test recognition, they need not be. The 132 inverted and paraphrased items (70 8Values, 62 Political Compass) were drafted by hand, checked against the automated double-negation and identity heuristics in `scoring/build_item_variants.py`, and reviewed before collection (`docs/item_variant_review.md`). Crossed with the same 18 models and both instruments at ten repetitions per cell, this yielded 1,080 administrations. The Contamination Index for a model-axis cell is the absolute score difference between the original and inverted conditions after un-flipping the inverted condition’s key, normalised by the instrument’s own score range; significance is a paired *t*-test per cell, Benjamini-Hochberg corrected across all 108 model-axis cells. A companion memorisation probe gave each of the 18 models the first half (by word count, minimum three words) of 20 items (10 per instrument, seeded selection) and asked for a verbatim completion, scored by sequence similarity to the true completion; 0.90 similarity is reported as “near-verbatim.” Two of the 18 models (`anthropic/claude-sonnet-4.5`, `nvidia/nemotron-3-ultra-550b-a55b`) returned a response the pipeline could not parse on this probe and are excluded from the rates reported below, which are therefore computed over 16 models.

Response format was realised as a fractional factorial over four response scales (the instrument’s own native scale, and three- and seven-point alternatives), two item orders (published, randomised), two option orders (published, reversed), and two elicitation modes (a forced label, and a free-text opinion subsequently classified onto the same scale by a fixed judge model, `openai/gpt-4o-mini`, held constant across every cell so judge identity is not a confound). This 24-cell design was crossed with the same 18 models at five repetitions per cell, yielding 4,320 administrations. Effect sizes are partial η^2 per axis from a five-factor ANOVA (model, scale, item order, option order, elicitation).

Each factorial was analysed on its own, against the unattributed, forced-choice baseline as the reference condition, rather than crossed with the other two factors and with instrument simultaneously; a single design crossing all four factors at once, at a cost proportional to their product rather than their sum, is future work.

A third instrument, SupplyValues, was administered to the same 18 models at its own native scale (46 items, three axes, -10 to $+10$ each), 30 trials per model, following the same lossless-port scoring approach as the other two instruments (`scoring/score_sapplyvalues.py`, ported from the instrument’s own open-source scoring code). Each model’s mean position is compared against the instrument’s own neutral point and, where an axis claims to share a construct with an existing instrument, against that axis’s model-level mean via Spearman correlation with a percentile bootstrap CI (2,000 resamples), following the same procedure as the bootstrap intervals in Section S1.6.

A fourth instrument, Pew Research Center’s 2026 Political Typology, was administered to the same

18 models, 10 trials per model (180 administrations). Pew’s methodology appendix for this typology describes its clustering procedure (weighted clustering around medoids) but does not publish the medoids, variable weights, or standardization constants needed to classify a new respondent independently of Pew’s own systems; reimplementing an approximate classifier from the general method description was therefore not possible, so, as with Political Compass, we drove the live 24-item public quiz (`scoring/score_pew_typology.py`) with each model’s answers and read off the classification Pew’s own site computes, rather than a score we derived ourselves. Item and option text (24 items across 21 navigation screens, 2–5 options each) was extracted directly from the live quiz and cross-checked against its own results-review page, which restates every item in full, disambiguated form. Because response options vary in wording by item, a model was asked to reply with the verbatim text of its chosen option per item rather than a shared code; the reply was matched to the live quiz’s option elements case- and punctuation-insensitively, with a whole-string-similarity fallback (≥ 0.85) for near-verbatim paraphrases. The outcome is each model’s classification into one of Pew’s nine typology groups, scored as that group’s 1–9 ordinal position in Pew’s own published left-to-right ordering.

Reusing the quiz’s item text and published group definitions for research is covered by Pew’s general terms of use, but submitting the interactive quiz programmatically 180 times (18 models $\times$ 10 trials) is a separate question from reusing the items themselves, and plausibly exceeds ordinary personal use of that interactive tool even though no respondent microdata or personal information is accessed at any point. We disclose this distinction explicitly rather than treating it as settled by the terms-of-use analysis above; see Section S15 and `docs/f3_instrument_terms_of_use.md` in the released repository for the full analysis.

Both instruments were then folded into the same crossed random-effects model used for the two-instrument budget in the article (`scoring/analyze_f3_variance_components.py`), rather than left as separate directional checks. Each instrument’s axis enters as one more trait level (`sapplyvalues:right`, `sapplyvalues:auth`, `sapplyvalues:prog`, `pew_typology:position`), z-scored within trait exactly as the two-instrument model requires, since the four instruments’ scales are mutually incompatible (± 10 , 0–100, ± 10 , and an ordinal 1–9 classification). Pew contributes a single trait, its one composite left-right placement, rather than per-axis scores like the other three, since that ordinal classification is Pew’s own output and not a score this project computed. Instrument identity remains folded into trait rather than fit as its own term, matching how the pooled two-instrument figure in the article is defined; separating instrument-level variance from axis-within-instrument variance is a further extension we do not attempt here. The two added instruments were collected for 18 of the 19 models (`mistralai/mistral-large-2512` was withdrawn from OpenRouter partway through collection), so the pooled panel is unbalanced by one model on those two traits; the crossed model does not require balance, and `results/f3_variance_components/f3_variance_components.json` records the composition explicitly.

S2 Extended Discussion

The finding that language models lean left of centre is real in the sense that it reproduces: we obtain it here on 19 models and two instruments, as others have obtained it elsewhere. What the article adds is a denominator. When the measured quantity is decomposed, the instrument contributes four times more variance than the model, and simple repetition of the same measurement contributes more than twice as much. The lean survives, but the *magnitude* of a reported lean is mostly not

about the model.

That distinction matters because the literature’s quantitative claims are magnitude claims. Statements that one model is more partisan than another, that a provider’s alignment procedure shifts a model’s politics by some amount, or that lean has grown or shrunk across model generations, all require that scores be comparable across studies. Our multi-trait multi-method result says they are not: the two most widely used instruments produce negatively correlated rankings of the same construct across the same models. Pooling or contrasting scores obtained with different instruments is therefore not a conservative approximation; it is an operation with no defined meaning.

We want to be precise about the scope of this claim, because it is easy to overstate. We are not arguing that political-bias audits are uninformative, nor that the models are politically neutral, nor that any particular published paper is wrong. Within a single study using a single instrument, model identity is clearly measurable and accounts for roughly two-thirds of the remaining variance; rank orderings obtained that way are meaningful. The failure is one of transportability. It also has a straightforward remedy, which is to report the apparatus alongside the estimate rather than the estimate alone.

That remedy has precedent in this journal specifically, applied to the same underlying algorithmic-fidelity assumption from a different angle. A study using ChatGPT to approximate American National Election Study responses finds the LLM’s synthetic responses show systematically *less* variance than the human responses they are meant to approximate, that minor prompt-wording changes shift the resulting distribution, and that the same prompt produces significantly different results three months apart (Bisbee et al. 2024). Our finding is the mirror image at a different level: even holding the prompt, the model, and the calendar date fixed, trial-to-trial variance within a single administration exceeds the variance between models. Both results caution against treating a single LLM output, or a small unexamined sample of them, as a valid measurement without first characterizing its error. A third study in this journal, evaluating LLMs’ zero-shot fidelity to human-written codebooks for political-text coding rather than questionnaire response, reaches a structurally similar conclusion for a different measurement task (Halterman and Keith 2025): across three independent lines of work on three different LLM measurement tasks, published in the same venue within about two years, none finds an off-the-shelf LLM output safe to treat as a measurement without characterizing how it fails.

Nor are we claiming that a questionnaire score predicts what a model would do given a real decision. A dual-instrument study comparing an abstract voting-advice questionnaire against real referendum outcomes finds that the two disagree: models cluster near the political centre on concrete votes despite a left-leaning profile on the questionnaire, and at least part of the questionnaire signal may reflect change-aversion rather than ideology (Barnettler 2026). That is a claim about behavioural validity, which this design does not test and does not need to in order for the instrument-comparability problem above to hold; an error budget for a questionnaire score is not an argument that the score predicts behaviour, and the two questions call for different studies. It is the reason a behavioural-validity arm, checking whether questionnaire scores predict downstream behaviour on concrete tasks, belongs in follow-up work rather than in this budget.

Our own earlier analysis is among the work this corrects. The stability metric we previously reported normalised dispersion by each sample’s own range, which assigns one of the worst stability scores in the cohort to a model whose answers varied by six hundredths of a point. We report that here (Section S7) rather than quietly replacing it, because the failure mode is not unique to us: any scale-free consistency statistic that divides by observed spread will invert in the same way on

tightly clustered responses, and several published consistency measures are constructed this way.

Three further components of the apparatus, each previously demonstrated only in isolation, turn out on direct measurement (Section S1.8) to be larger than the model term in the article’s error budget, not smaller. Accommodation to the inferred asker (Törnberg and Schimmel 2026) explains more variance ($\eta^2 = 0.79$) than model identity does in the same unattributed design ($\eta^2 = 0.56$ – 0.73); free-text versus forced-choice elicitation (Röttger et al. 2024) explains more variance (39%) than model identity within the same format factorial (23%); and 17 of 18 models show a measurable, FDR-corrected score shift under item inversion, consistent with instrument recognition rather than pure engagement with item content (Bianchi et al. 2026). None of these three is folded into the pooled error budget, because each was collected as its own factorial against the unattributed, forced-choice baseline rather than crossed with the other two and with instrument simultaneously; a single design that crosses all four factors at once is the natural next step, and would very likely drive the model term’s already-small pooled share (13%, itself an overlapping upper bound with any one of these factors held at its literature-typical, unattributed, forced-choice default) down further still. That is the direction every measurement factor examined so far has pushed, and none has pushed the other way.

A minimum reporting standard for political-bias audits

On the evidence above, we suggest that a political-bias audit of a language model should report the following alongside any score. None of these is expensive; together they make an estimate interpretable and comparable.

1. **The instrument, named, with its items.** Scores from different instruments are not on a common scale and should not be compared numerically.
2. **The asker condition.** State the framing presented to the model, including the absence of one. An unattributed prompt is a condition, not a neutral baseline.
3. **Repetition count and dispersion.** Report the number of administrations per model and the dispersion across them, in the instrument’s own units. Trial noise here exceeded the model effect.
4. **An error budget, or at minimum a variance decomposition** across whatever measurement factors were varied. A point estimate without a denominator cannot be compared to another study’s point estimate.
5. **A contamination check** for instruments that appear in pretraining corpora, such as an item-inversion or paraphrase control.
6. **Item-level results**, or at least the dispersion across items. Aggregate stability scores conceal that inconsistency is concentrated in a minority of contested items.

Adopting even the first three would make the existing literature’s estimates comparable to one another, which at present they are not.

S3 Directional Consistency, with Substantial Magnitude Differences

Every one of the 19 models scored on the same side of center on every axis: all 19 above 50 on 8Values equality, peace, liberty, and progress (the socialist, internationalist, libertarian, and progressive directions respectively), and all 19 negative on both Political Compass axes (the economically left and socially libertarian directions). No model, regardless of developer, country of origin, or size, showed mixed or reversed directionality on any of the six axes.

What varies enormously is the size of the gaps between models. On 8Values equality, means range from 60.91 (x-ai/grok-4.20) to 83.08 (qwen/qwen3-30b-a3b); on 8Values liberty, from 57.80 (deepseek/deepseek-chat) to 71.81 (mistralai/mistral-large-2512); on 8Values progress, from 64.47 (deepseek/deepseek-chat) to 82.97 (mistralai/mistral-large-2512); on Political Compass economic, from -8.09 (mistralai/mistral-large-2512) to -4.89 (anthropic/claude-sonnet-4.5); and on Political Compass social, from -7.82 (google/gemini-2.5-flash) to -5.15 (qwen/qwen3-30b-a3b). Table S1 reports the full per-model mean on every axis, and Figures S1 and S2 show the same means plotted directly: every model occupies the same quadrant of the Political Compass plane, and every model sits on the same side of the neutral reference line on all four 8Values axes, with the between-model spread visible at a glance rather than only in tabular form.

Table S1: Mean score by model and axis (8Values axes 0–100; Political Compass axes -10 to 10).

Model	Equality	Peace	Liberty	Progress	Econ.	Social
amazon/nova-pro-v1	69.95	66.21	62.50	66.16	-6.26	-5.30
anthropic/claude-haiku-4.5	71.82	70.21	64.33	68.26	-6.85	-6.53
anthropic/claude-opus-4.5	69.92	70.57	68.11	77.62	-6.44	-7.44
anthropic/claude-sonnet-4.5	71.05	66.57	63.85	71.89	-4.89	-6.27
cohere/command-r-plus	72.19	69.49	65.58	75.03	-5.32	-5.76
deepseek/deepseek-chat	67.52	63.45	57.80	64.47	-6.30	-5.43
deepseek/deepseek-v3.2	69.84	67.77	62.36	69.91	-6.18	-5.86
google/gemini-2.5-flash	72.33	72.26	69.68	78.16	-7.94	-7.82
meta-llama/llama-3.3-70b	68.76	69.75	65.38	66.38	-6.31	-6.29
meta-llama/llama-4-maverick	75.28	75.51	68.47	73.60	-6.03	-6.14
mistralai/mistral-large	80.08	74.54	71.81	82.97	-8.09	-7.45
mistralai/mistral-small	69.23	69.57	64.36	79.21	-7.27	-6.55
nvidia/nemotron-3-ultra	63.01	62.52	60.98	65.47	-6.04	-6.96
openai/gpt-4o	73.48	74.77	70.51	82.44	-6.44	-7.34
openai/gpt-4o-mini	76.51	72.97	67.49	80.94	-5.32	-6.65
openai/gpt-5-mini	73.28	69.20	65.48	73.20	-6.04	-6.44
qwen/qwen3-235b	79.81	75.34	70.43	77.58	-6.55	-6.26
qwen/qwen3-30b	83.08	75.08	69.42	78.83	-6.12	-5.15
x-ai/grok-4.20	60.91	67.03	69.38	78.98	-7.19	-7.77

Political Compass position of all 19 models (mean of 60 trials each); + marks the neutral human-baseline proxy

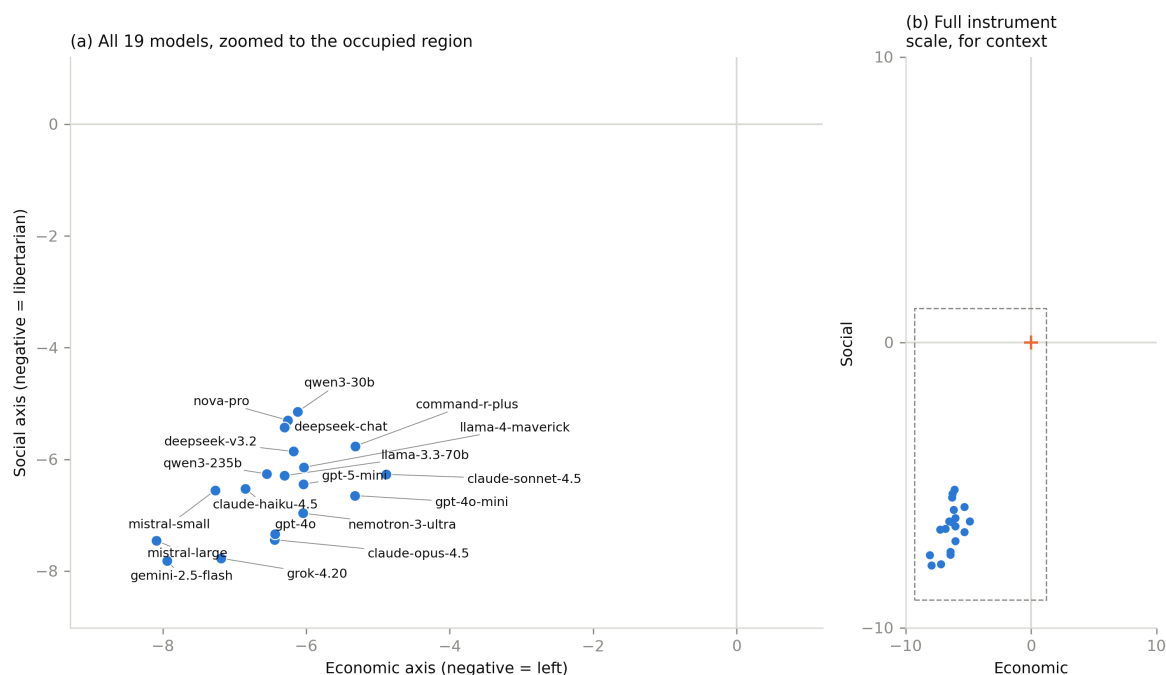


Figure S1: Political Compass position of all 19 models (mean of 60 trials each). All models fall in the economically left, socially libertarian quadrant.

8Values scores of all 19 models, four axes (mean of 60 trials each)

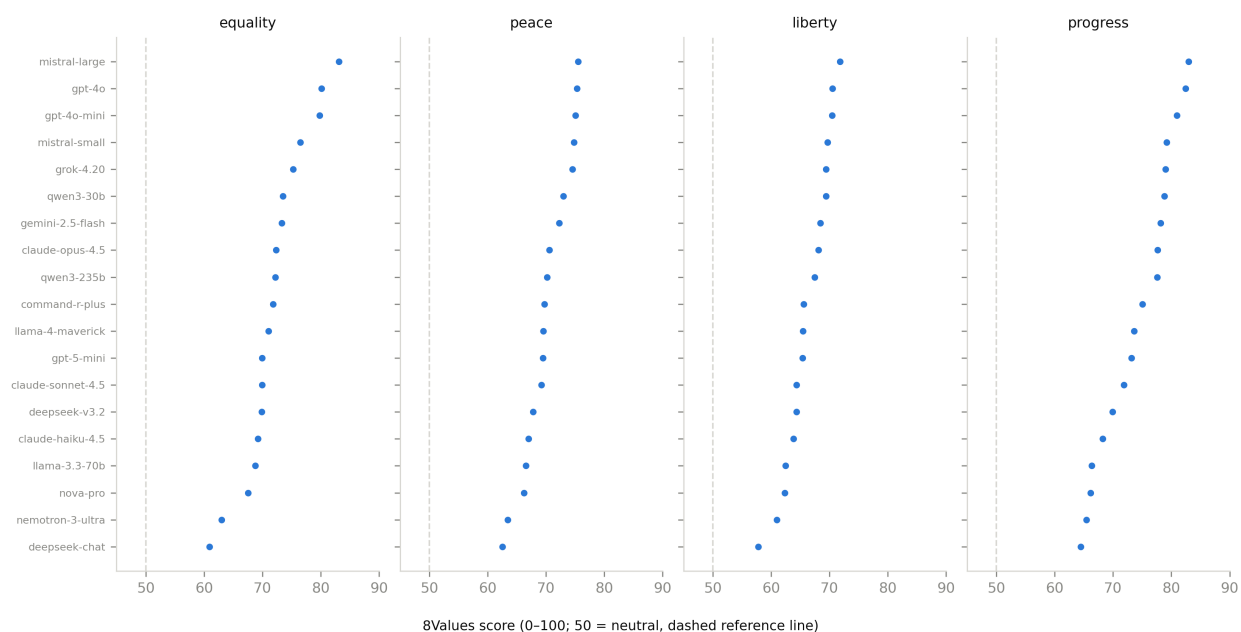


Figure S2: 8Values scores of all 19 models on all four axes (mean of 60 trials each), sorted independently within each panel; the dashed line marks the instrument's neutral center-point (50).

S4 Omnibus Tests: Large, Robust Effects on Every Axis

Table S2 reports the classical one-way ANOVA, Welch’s ANOVA, and the Kruskal-Wallis test, together with the omnibus eta-squared effect size, for each of the six scored axes. All three tests agree at $p < .0001$ on every axis, and every effect size is large (η^2 between 0.5625 and 0.7263). Shapiro-Wilk rejects normality for a majority of the 19 groups on most axes (11 to 17 of 19 groups non-normal, depending on axis), and Levene’s test rejects equal variance on every axis ($p < .0001$ throughout), which is why Welch’s ANOVA and Kruskal-Wallis, not the classical ANOVA, are treated as the primary evidence for these omnibus effects.

Table S2: Omnibus tests across all 19 models, by axis. All three tests agree at every axis.

Test	Axis	Classical F	Welch F	Kruskal-Wallis H	η^2
8Values	Equality	138.3	534.8	792.0	0.6896
8Values	Peace	80.1	421.5	759.4	0.5625
8Values	Liberty	89.3	459.2	753.5	0.5891
8Values	Progress	165.3	618.5	886.8	0.7263
Pol. Compass	Economic	124.8	2240.3	762.0	0.6670
Pol. Compass	Social	163.3	1370.3	867.5	0.7239

Note: all $p < .0001$ for all three tests, on every axis.

S5 Post-hoc Comparisons: Games-Howell as the Primary Test, Corrected for the Full Test Family

With 19 groups, each axis has 171 pairwise comparisons, and six axes give a full family of 1,026 pairwise tests reported across this paper (`scoring/analyze_posthoc.py`; `results/posthoc/posthoc_results.json`). Because equal variances are rejected on every axis, Games-Howell is treated as primary, with Tukey HSD reported alongside. At raw $p < 0.05$, Games-Howell finds between 121 and 144 of the 171 pairs significant per axis (equality: 142; peace: 136; liberty: 131; progress: 144; economic: 121; social: 138), 812 of the 1,026 pairwise tests in total. Applying a Benjamini-Hochberg false discovery rate correction across the full family of 1,026 tests, rather than within each axis separately, leaves 809 of the 812 significant, a negligible change (equality: 142/142; peace: 136/136; liberty: 131/131; progress: 142/144; economic: 120/121; social: 138/138): the between-model differences reported here are not an artifact of running many uncorrected tests. The two post-hoc tests mostly agree with each other as well: wherever Tukey finds a different result than raw- p Games-Howell, only 5 to 9 pairs per axis are significant under Tukey but not under Games-Howell, an artifact of Tukey’s equal-variance assumption rather than a real disagreement about the larger pattern.

Several pairwise Games-Howell effect sizes (Hedges’ g) on this dataset are very large in magnitude, considerably larger than is typical for behavioral data. This is not evidence of unusually large real differences so much as a consequence of some models, particularly several Claude variants, having extremely low within-model variance on some axes (Section S7): dividing a modest mean difference by a very small pooled standard deviation produces a large standardized effect size even when the raw difference, on each instrument’s own -10 -to- 10 or 0 -to- 100 scale, is unremarkable. Readers should interpret the reported Hedges’ g values alongside the raw mean differences in Table S1, not in isolation.

S6 Item-Level Instability is Concentrated in a Minority of Items

Aggregating consistency to the axis level, as Section S7 does, hides where inconsistency actually lives. Measured per item, 44.8% of model–item cells are perfectly stable across all 60 repetitions, and the most volatile decile of items carries only 14.4% (8Values) and 15.9% (Political Compass) of that instrument’s total response entropy (Figure S3). Instability is a property of particular contested items, not a global trait of a model, and a model-level or axis-level stability score averages the two together. This is the reason recommendation 6 of the minimum reporting standard (Section S2) asks for item-level results specifically, rather than an axis-level summary alone.

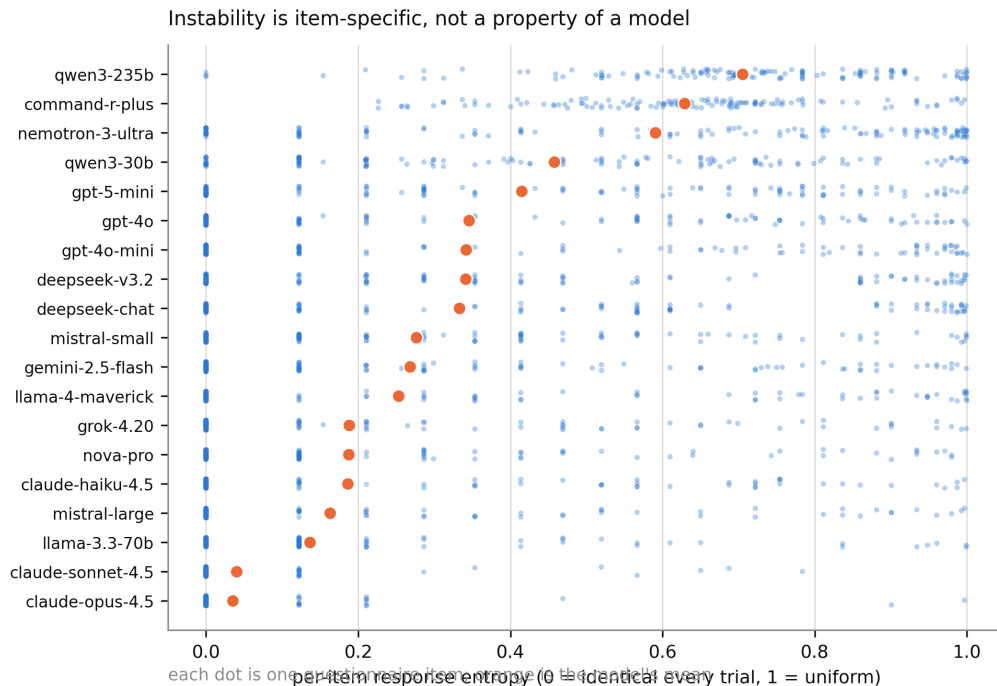


Figure S3: Per-item response entropy by model. Each point is one questionnaire item; orange marks the model mean. 44.8% of model–item cells are perfectly stable across all 60 repetitions, and the most volatile decile of items carries under 16% of total entropy, so inconsistency is concentrated in particular contested items rather than distributed across a model’s responses.

S7 Consistency: Range-Normalized Stability Varies by Model, not by Provider

Ranking every model-axis combination by ISS (Section S1.6) shows that the most and least consistent entries in the dataset are not concentrated in any one provider, country, or size tier. Table S3 lists the ten most and ten least consistent combinations out of all 114 (19 models $\times$ 6 axes).

Table S3: Ten most and ten least consistent model-axis combinations by Ideological Stability Score (ISS). Lower ISS = more consistent.

Model (test/axis)	ISS
<i>Most consistent</i>	
anthropic/claude-opus-4.5 (Pol. Compass/social)	0.00
anthropic/claude-opus-4.5 (8Values/peace)	7.62
anthropic/claude-opus-4.5 (8Values/equality)	7.87
anthropic/claude-sonnet-4.5 (Pol. Compass/social)	8.26
meta-llama/llama-3.3-70b (8Values/equality)	8.33
anthropic/claude-sonnet-4.5 (Pol. Compass/economic)	9.04
amazon/nova-pro-v1 (Pol. Compass/economic)	9.21
x-ai/grok-4.20 (8Values/peace)	10.20
meta-llama/llama-3.3-70b (8Values/peace)	12.11
meta-llama/llama-3.3-70b (8Values/progress)	12.92
<i>Least consistent</i>	
qwen/qwen3-235b (8Values/equality)	34.14
nvidia/nemotron-3-ultra (8Values/liberty)	35.58
meta-llama/llama-4-maverick (8Values/liberty)	36.19
deepseek/deepseek-chat (8Values/equality)	37.45
google/gemini-2.5-flash (8Values/peace)	37.71
nvidia/nemotron-3-ultra (8Values/peace)	40.50
nvidia/nemotron-3-ultra (8Values/progress)	41.39
nvidia/nemotron-3-ultra (8Values/equality)	42.76
anthropic/claude-haiku-4.5 (Pol. Compass/economic)	44.42
anthropic/claude-opus-4.5 (Pol. Compass/economic)	65.22

Table S3 shows only the twenty most extreme entries; Figure S4 shows the full distribution of all 114 ISS values, grouped by axis, and makes visible both the general spread within each axis and the single high-leverage outlier (anthropic/claude-opus-4.5’s Political Compass economic score) discussed below.

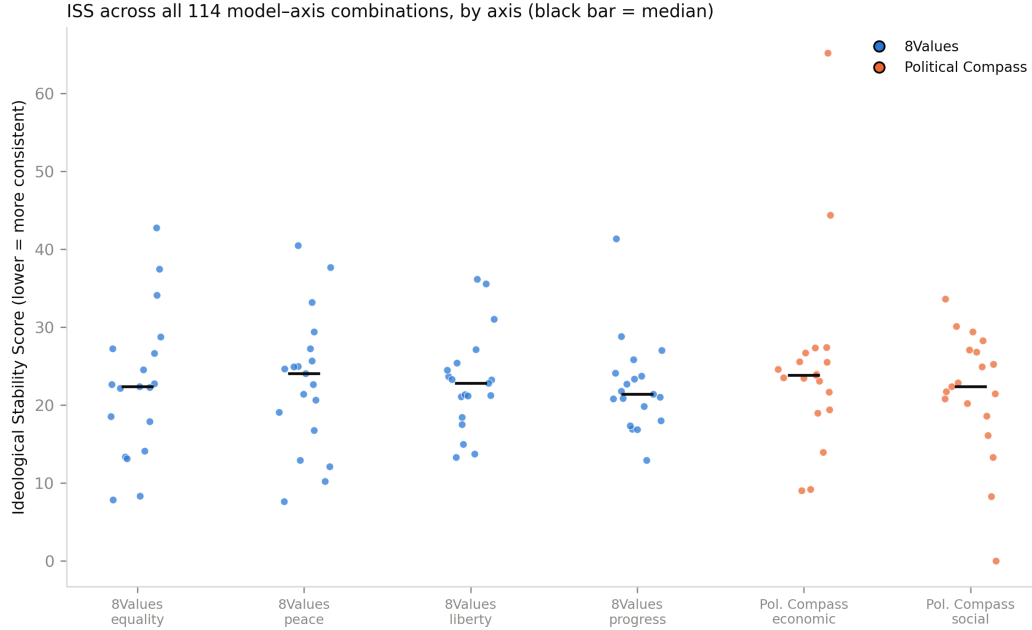


Figure S4: Ideological Stability Score for all 114 model–axis combinations, grouped by axis. Each point is one model; the black bar marks the axis median.

anthropic/claude-opus-4.5 and meta-llama/llama-3.3-70b-instruct each appear three times among the ten most consistent combinations, more than any other model, while nvidia/nemotron-3-ultra-550b-a55b appears four times among the ten least consistent. No provider is exclusively at either extreme: Claude variants appear among both the single most consistent entry and, on a different axis, the single least consistent entry in the entire dataset.

That last case illustrates the ISS range-normalization caveat from Section S1.6 directly, and is the reason the article reports dispersion in the instrument’s own units and the residual variance component of the error budget instead of ISS. The Political Compass economic score of anthropic/claude-opus-4.5 across its 60 trials took only two distinct values, -6.50 and -6.38 , an absolute range of 0.12 on a -10 -to- 10 scale; normalizing by that tiny self-range produces the highest ISS in the dataset (65.22) despite the underlying answers being, in absolute terms, essentially constant. By contrast, the same model’s Political Compass social score was literally identical across all 60 trials (-7.44 every time, $\text{ISS} = 0.00$). Both figures describe a model giving highly stable absolute answers; the ISS score for the economic axis should be read as an artifact of the metric’s normalization, not as evidence of erratic behavior. anthropic/claude-haiku-4.5’s economic-axis ISS (44.42) reflects a genuinely larger absolute range (-7.38 to -6.38) and is a more straightforward instance of comparatively lower consistency. Across all 114 model–axis cells, ISS correlates with absolute dispersion (standard deviation) at only $\rho = 0.46$ (Figure S5), which is the quantitative version of the same point: ISS is not, in general, a good proxy for how much a model’s answers actually vary.

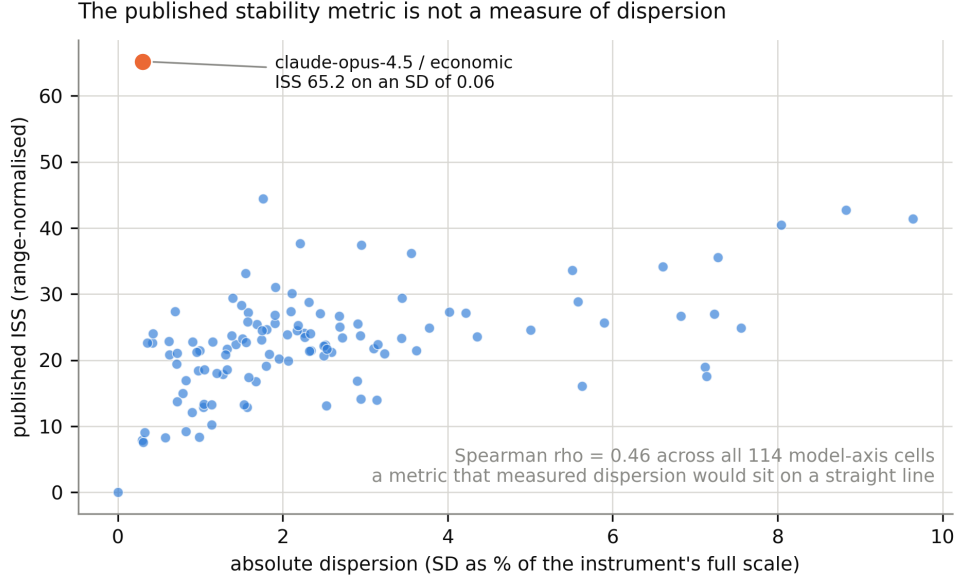


Figure S5: The range-normalised stability metric is not a measure of dispersion. Each point is one model-axis cell. Normalising by each sample’s own observed range inflates the score for models whose answers are tightly clustered: the highlighted cell registers 65.2 on a standard deviation of 0.060. Across all 114 cells the metric correlates with absolute dispersion at only $\rho = 0.46$.

S8 Comparison Against a Human-Population Reference Point

Beyond comparing models to each other, every one of the 114 model-axis combinations (19 models $\times$ 6 axes) differs significantly from that instrument’s own neutral center-point (one-sample t -test against 0 for the two Political Compass axes and against 50 for the four 8Values axes), and this holds for all 114 tests even after a Benjamini-Hochberg false discovery rate correction. Given that Gallup’s 2024 Values and Beliefs poll finds the US public close to an even three-way split between conservative, moderate, and liberal self-identification on social issues, and only moderately right-of-center on economic issues (Gallup 2024), a neutral center-point is a reasonable, if bounded, proxy for a human population reference (Section S1.6). Every model in this study, without exception, differs significantly from that proxy, in the same direction identified in the between-model comparisons above: left of center economically and libertarian of center socially.

Statistical significance alone is a weak claim at $n = 60$ trials per test, since even a small, practically unremarkable deviation from center would clear $p < 0.05$ at that sample size. The effect sizes rule this out: computing a one-sample Cohen’s d for each of the 114 tests (the standardized deviation of a model’s mean from the instrument’s center-point) finds every finite value at $|d| \geq 1.47$, comfortably past Cohen’s conventional threshold for a “large” effect ($|d| \geq 0.8$) (Cohen 1988), with a median $|d|$ of 11.92 and a maximum of 106.74 (one model-axis combination is excluded from these summary statistics for having an effectively zero within-model standard deviation, discussed in Section S7). The deviation from the neutral center-point is therefore not merely statistically detectable but, on average, an order of magnitude larger than the conventional threshold for a large effect.

S9 Prompt-Robustness Check: Magnitude Shifts, Direction Does Not

Across the five-model, four-variant check (Section S1.7), prompt wording produced a statistically significant shift in 29 of 30 model-axis combinations tested (one-way ANOVA across the five prompt conditions, $p < 0.05$), and the effect size was large ($\eta^2 \geq 0.14$) in 28 of those 30, with η^2 ranging from 0.02 to 0.75 (median 0.37). This is a genuine, non-trivial finding, not a null result: instructional-wrapper wording measurably moves scores for most models on most axes, consistent with Röttger et al.’s critique that forced-choice political-typology evaluations are sensitive to exactly how the task is framed (Röttger et al. 2024).

What did not move is direction. Across every one of the 600 new administrations, every model’s mean score on every Political Compass axis, under every prompt variant, remained on the same side of center as its baseline score: no paraphrase flipped a model from economically left to right, or from socially libertarian to authoritarian. The largest single shift observed was `deepseek/deepseek-chat`’s 8Values equality score, ranging from 66.45 to 78.19 (an 11.7-point spread) across the five prompt conditions, still comfortably on the equality (left) side of the 50-point center in every condition. The directional finding that every model leans the same way on every axis is therefore robust to prompt wording; the exact magnitude of that lean, and consequently any fine-grained ranking between closely-scored models, should be treated as sensitive to the specific wording used and not over-interpreted.

S10 Open-Ended-Generation Validation Arm

Across the 120 judged generations (Section S1.7), the correlation between a model’s mean Political Compass score and its mean judge-rated lean when writing open-endedly about matched-axis topics was weak and not statistically significant on either axis (economic Pearson $r = -0.56$, $p = 0.32$; social $r = 0.27$, $p = 0.67$; $n = 5$ models). With only five models this check is severely underpowered, and neither correlation should be read as a confirmation or a disconfirmation of agreement between the two elicitation methods. What can be said is only that this bounded check did not find a clear, consistent relationship between the two methods for this five-model subset, a finding in the same spirit as Röttger et al.’s broader point that forced-choice and open-ended elicitation can diverge (Röttger et al. 2024), and one that argues for a larger-scale version of this arm (more models, more topics) before drawing a firm conclusion either way.

S11 Item-Response-Theory Reanalysis

The published Political Compass and 8Values scoring keys treat every item as equally informative and cannot separate a model declining to engage from a model holding a genuinely moderate position (Wachter et al. 2025). We refit a graded-response IRT model separately on each of the six axes (the four 8Values axes and the two Political Compass axes), using every item-level response in the baseline dataset (19 models $\times$ 60 trials), and compare the resulting latent trait (θ) against each model’s published axis score.

The published scores track the IRT-derived positions closely on five of the six axes.

Spearman correlations between θ and the published score range from $\rho = 0.86$ to $\rho = 0.94$ ($p < 10^{-5}$ in every case) on 8Values equality, peace, liberty, and progress and on the Political Compass social axis, with narrow Bland-Altman limits of agreement (Table S4). **The exception is the Political Compass economic axis**, where $\rho = 0.64$ and the limits of agreement are roughly twice as wide as on any other axis. That axis has only 11 items, the fewest of the six, and half its information comes from just 2 of them (Table S4). The published economic score for this instrument rests on the fewest and least-distributed items in the study, which is a specific, checkable reason to weight it less heavily than the other five axes rather than a general indictment of the scoring approach.

Information is concentrated in a minority of items on every axis. Between 4 and 12 items (out of 11 to 50 per axis) carry half of each axis’s total discrimination, and the top decile of items by discrimination carries 24–33% of it. A small, identifiable subset of items is doing most of the work of separating models on every axis we fit, on both instruments.

Neutral-response rates vary roughly six-fold across models on 8Values (9% to 52% of items answered at the scale midpoint), which is the avoidance behaviour Wachter et al. (2025) predict: at least part of the apparent centrism in the most avoidant models is a refusal to commit to either pole rather than a moderate position, and it is not evenly distributed across the cohort. We do not have a validated way to convert a neutral-rate estimate into a bias correction for the published score, so we report it as a per-model caveat rather than an adjustment.

Table S4: IRT reanalysis summary by axis. ρ is the Spearman correlation between the IRT latent trait and the published axis score across the 19 models; LoA is the Bland-Altman 95% limits of agreement, in the published score’s own units after within-axis standardization; “items for half info.” is the minimum number of items whose summed discrimination reaches half the axis total.

Instrument	Axis	Items	ρ	LoA	Items for half info.
8Values	Equality	22	0.91	± 0.83	4
8Values	Peace	24	0.86	± 0.88	6
8Values	Liberty	37	0.92	± 0.61	7
8Values	Progress	33	0.88	± 0.81	7
Political Compass	Economic	11	0.64	± 1.27	2
Political Compass	Social	50	0.94	± 0.63	12

S12 Multi-Trait Multi-Method Matrix

Figure S6 shows the full correlation matrix behind the convergent-validity result reported in the article. Convergent coefficients (same construct, different instrument) average $r = -0.26$; discriminant coefficients (different construct, same instrument) average $r = +0.52$ (Table S5). The four 8Values axes correlate with one another at mean $r = 0.71$, consistent with a single dominant general factor rather than four separable traits.

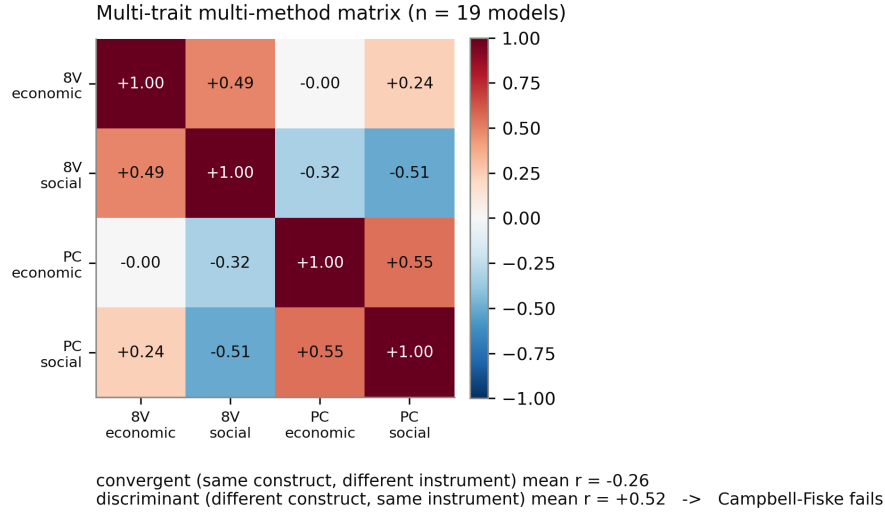


Figure S6: Multi-trait multi-method matrix across the 19 models. Convergent coefficients (same construct, different instrument) average $r = -0.26$; discriminant coefficients (different construct, same instrument) average $r = +0.52$. The Campbell–Fiske ordering is inverted, so the two instruments cannot be treated as interchangeable measures of a common political position.

Table S5: Multi-trait multi-method coefficients across the 19 models. Convergent validity requires that the same construct measured by two instruments correlate more strongly than two different constructs measured by the same instrument. It does not.

Type	Pair	r
Convergent	8values:economic vs political_compass:economic	-0.005
Convergent	8values:social vs political_compass:social	-0.506
Discriminant	8values:economic vs 8values:social	+0.490
Discriminant	political_compass:economic vs political_compass:social	+0.549
Mean convergent		-0.255
Mean discriminant		+0.520

A preliminary check on a third instrument, SupplyValues, is consistent with this picture rather than resolving it. The directional finding partly replicates: all 18 models scored are economically left ($\text{right} < 0$) and culturally progressive ($\text{prog} > 0$), matching the Political Compass and 8Values baseline; the libertarian finding does not, with only 7 of 18 models scoring libertarian on SupplyValues’ authority axis. Correlating each SupplyValues axis against the existing axis that claims to measure the same construct, across the 18 shared models, the economic axis disagrees with Political Compass ($\rho = -0.49$, 95% CI $[-0.78, -0.04]$, bootstrap, negative where the shared “positive means right” convention predicts positive) while agreeing with 8Values’ equality axis in the expected direction ($\rho = -0.53$, $[-0.87, -0.01]$). The authority-axis comparisons are inconclusive at this sample size (95% CIs spanning zero against both other instruments). We read this as a third data point for the same conclusion, not a resolution of it: adding an instrument did not converge the disagreement onto a single answer, it added a third position.

A fourth instrument, Pew Research Center’s 2026 Political Typology, gives an authoritative check

of a different kind (Section S1.8): rather than scoring an axis ourselves, we drove the live 24-item quiz with each model’s answers and read off the classification Pew’s own site computes, since Pew’s clustering weights are not published. Sixteen of 18 models’ modal classification falls left of centre (one of the four left-leaning groups in Pew’s own nine-group, left-to-right ordering); two, `deepseek/deepseek-chat` and `x-ai/grok-4.20`, classify as *Pragmatic and Polite Right*, the first group right of centre, with `x-ai/grok-4.20` landing there in all 10 of 10 trials. The ordinal position of a model’s classification (1–9, left to right) does not correlate with its Political Compass economic score or 8Values equality score across the 18 shared models ($\rho = 0.04$, $[-0.39, 0.51]$, and $\rho = -0.03$, $[-0.55, 0.54]$ respectively, both consistent with no relationship at this sample size). The directional finding therefore replicates on a fourth, independently classified instrument, with two clear exceptions.

The four-instrument crossing reported in the article (model-identity share 1.89%; SapplyValues alone, 0.49%; Pew alone, 8.21%) is the full accounting that this section’s pairwise checks anticipate. Between-model spread and within-model trial noise are proportionally in line with the other instruments for SapplyValues too (ratio ≈ 1.9 for all three continuous instruments), so the collapse is not a SapplyValues data-quality artefact; it is what the objection “a variance component estimated from two instruments is meaningless” looks like quantitatively once a third and fourth instrument are added.

S13 Display Items Not Referenced Elsewhere

Your Political Compass

Economic Left/Right: -6.38

Social Libertarian/Authoritarian: -5.85

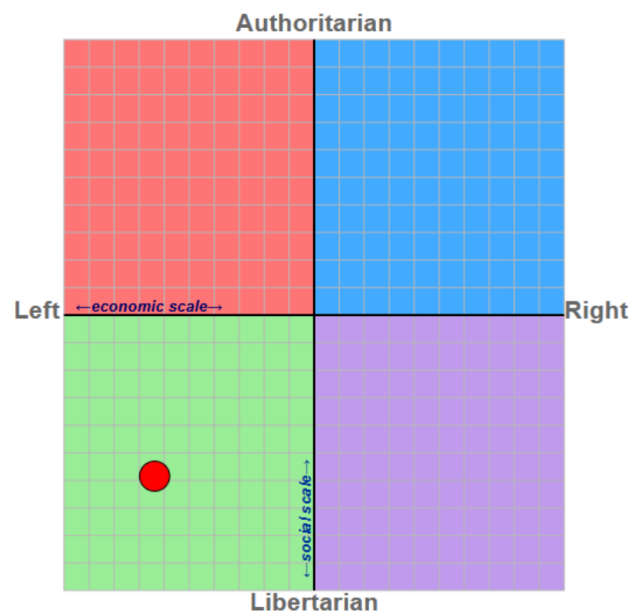


Figure S7: Example Political Compass output for a single trial.

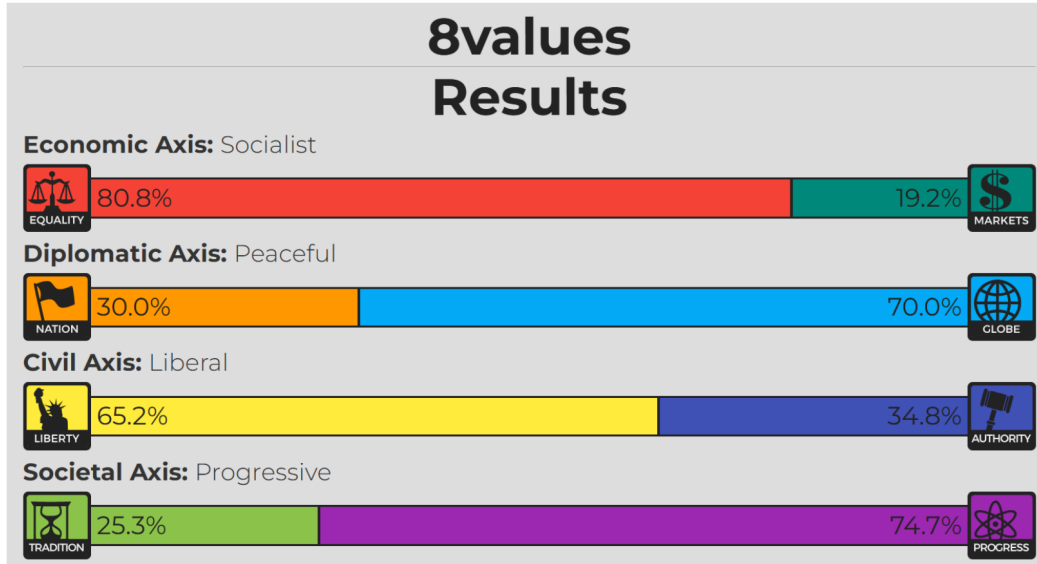


Figure S8: Example 8Values output for a single trial.

Table S6: Collection completion rate and cost by model (both tests combined, 120 administrations/model).

Model	Completion rate	Cost
amazon/nova-pro-v1	100%	\$0.31
anthropic/claude-haiku-4.5	100%	\$0.42
anthropic/claude-opus-4.5	100%	\$1.95
anthropic/claude-sonnet-4.5	100%	\$1.42
cohere/command-r-plus-08-2024	100%	\$0.98
deepseek/deepseek-chat	100%	\$0.09
deepseek/deepseek-v3.2	100%	\$0.02
google/gemini-2.5-flash	100%	\$0.21
meta-llama/llama-3.3-70b-instruct	100%	\$0.04
meta-llama/llama-4-maverick	100%	\$0.07
mistralai/mistral-large-2512	100%	\$0.11
mistralai/mistral-small-3.2-24b-instruct	100%	\$0.02
nvidia/nemotron-3-ultra-550b-a55b	100%	\$1.38
openai/gpt-4o	100%	\$0.84
openai/gpt-4o-mini	100%	\$0.04
openai/gpt-5-mini	100%	\$0.59
qwen/qwen3-235b-a22b	100%	\$0.49
qwen/qwen3-30b-a3b	100%	\$0.14
x-ai/grok-4.20	100%	\$0.18
Total	100% (2,280/2,280)	\$9.30

S14 Data and Code Availability

All raw per-trial data from this study are released at <https://github.com/sletch1/AI-Political-Bias-Research-Paper>. This includes the main 2,280-administration run across 19 models

(`data/raw_trials/`, with a flattened table in `data/scores.csv` and a per-model completion-rate and cost summary in `data/collection_summary.md`); the 2,160-administration asker-identity factorial and its 180-call probe (`data/asker_identity/`); the 1,080-administration contamination factorial and its memorisation probe (`data/item_variants/`); the 4,320-administration response-format factorial (`data/format_factorial/`); a 540-administration baseline on a third instrument, SupplyValues (18 models, `data/raw_trials/`, filenames containing `sapplyvalues`); and a 180-administration baseline on a fourth instrument, Pew’s 2026 Political Typology (`data/pew_typology/`). Both the third and fourth instrument are crossed into the shared variance-components model used for the first two (`scoring/analyze_f3_variance_components.py`; Section S1.8). Every record includes the model’s structured JSON answers, the resulting score, and the exact API cost of that call.

The complete pipeline, including the collection script, both scoring adapters (the 8Values port and the Political Compass browser-automation adapter), the consolidation script, the figure-generation script, and the full statistical analysis script, is released as open source at the same repository. No code is reproduced in the body of this paper; the repository is the authoritative source for exact implementation details and is kept in sync with the analyses reported here. A snapshot of the data and code at submission time is additionally archived at <https://doi.org/10.5281/zenodo.22968158>.

S15 Declarations

Author Contributions

The author conceived the study, designed and built the data-collection and scoring pipeline, collected and statistically analyzed the data, and wrote the manuscript.

Competing Interests

The author declares no competing interests.

Funding

The author received no specific funding for this work.

Acknowledgments: Use of AI Tools

Portions of this manuscript’s text were drafted and revised with the assistance of a commercial AI writing assistant, used interactively by the author throughout the project’s development in 2026, accessed via a paid subscription. See Section S1.5 for the corresponding disclosure of AI-assisted code and analysis. The author directed all content decisions, verified all reported numbers against the underlying data and code, and takes full responsibility for the manuscript’s content and claims.

Ethics Declarations

This study did not involve human participants, human data, or animal subjects. All data consist of text outputs generated by third-party large language models accessed via a commercial API (OpenRouter); no personal, sensitive, or identifiable information is involved. Institutional ethics review was therefore not required.

One instrument, Pew Research Center’s 2026 Political Typology (Section S1.8), was scored by programmatically driving Pew’s live public quiz and reading off its own classification, rather than reimplementing Pew’s unpublished clustering procedure, following the same approach already used for Political Compass. Reusing the instrument’s item text and published group definitions for research is covered by Pew’s general terms of use. Submitting the interactive quiz itself 180 times in an automated fashion is a distinct question: it plausibly exceeds ordinary personal use of that interactive tool, even though it accesses no respondent microdata or personal information at any point. We disclose this explicitly rather than treating it as settled by the terms-of-use analysis for the survey items.

References

- 8Values (2019). 8values. Github.io.
- Anthropic (2023). Claude. Claude.ai.
- Argyle, L. P., E. C. Busby, N. Fulda, J. R. Gubler, C. Rytting, and D. Wingate (2023). Out of one, many: Using language models to simulate human samples. *Political Analysis* 31(3), 337–351.
- Barnettler, J. P. (2026). Progressive in principle, centrist in practice: LLM political bias is instrument-dependent.
- Bianchi, B., D. Tiscornia, M. Travizano, and A. Futoransky (2026). Political plasticity: An analysis of ideological adaptability in large language models.
- Bisbee, J., J. D. Clinton, C. Dorff, B. Kenkel, and J. M. Larson (2024). Synthetic replacements for human survey data? the perils of large language models. *Political Analysis* 32(4), 401–416.
- Buyl, M., A. Rogiers, S. Noels, I. Dominguez-Catena, E. Heiter, R. Romero, I. Johary, A.-C. Mara, J. Lijffijt, and T. De Bie (2025). Large language models reflect the ideology of their creators. *npj Artificial Intelligence* 2, 7.
- Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Lawrence Erlbaum Associates.
- DeepSeek (2025). Deepseek.
- Gallup (2024). Increase in liberal views brings ideological parity on social issues. Gallup Values and Beliefs poll, May 1–23, 2024, n=1,024 U.S. adults.
- Hagendorff, T. (2023). Machine psychology: Investigating emergent capabilities and behavior in large language models using psychological methods. arXiv.org.
- Halterman, A. and K. A. Keith (2025). Codebook LLMs: Evaluating LLMs as measurement tools for political science concepts. *Political Analysis*.

- Harris, C. R., K. J. Millman, S. J. van der Walt, R. Gommers, P. Virtanen, D. Cournapeau, E. Wieser, J. Taylor, S. Berg, N. J. Smith, R. Kern, M. Picus, S. Hoyer, M. H. van Kerkwijk, M. Brett, A. Haldane, J. F. del Río, M. Wiebe, P. Peterson, and P. Gérard-Marchant (2020). Array programming with NumPy. *Nature* 585(7825), 357–362.
- Miotto, M., N. Rossberg, and B. Kleinberg (2022). Who is GPT-3? an exploration of personality, values and demographics. In *Proceedings of the 5th Workshop on Natural Language Processing and Computational Social Science (NLP+CSS), co-located with EMNLP 2022*, pp. 218–227.
- OpenAI (2023). Chatgpt.
- Röttger, P., V. Hofmann, V. Pyatkin, M. Hinck, H. R. Kirk, H. Schütze, and D. Hovy (2024). Political compass or spinning arrow? towards more meaningful evaluations for values and opinions in large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL 2024)*.
- Seabold, S. and J. Perktold (2010). Statsmodels: Econometric and statistical modeling with python. In *Proceedings of the 9th Python in Science Conference*.
- Serapio-García, G., M. Safdari, C. Crepy, L. Sun, S. Fitz, P. Romero, et al. (2023). Personality traits in large language models. arXiv.org.
- The Political Compass (n.d.). The political compass.
- Törnberg, P. and M. Schimmel (2026). Political bias audits of LLMs capture sycophancy to the inferred auditor.
- Tukey, J. W. (1949). Comparing individual means in the analysis of variance. *Biometrics* 5(2), 99–114.
- Virtanen, P., R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, and C. J. Carey (2020). SciPy 1.0: fundamental algorithms for scientific computing in python. *Nature Methods* 17(3), 261–272.
- Wachter, J., M. Radloff, M. Smolej, and K. Kinder-Kurlanda (2025). Are LLMs (really) ideological? an IRT-based analysis and alignment tool for perceived socio-economic bias in LLMs.